

Question ID 93f48423

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 93f48423

What is the area, in square inches, of a rectangle with a length of **7** inches and a width of **6** inches?

- A. **13**
- B. **20**
- C. **42**
- D. **84**

ID: 93f48423 Answer

Correct Answer:

C

Rationale

Choice C is correct. The area, ***A***, of a rectangle is given by the formula **$A = \ell w$** , where ***ℓ*** represents the length of the rectangle and ***w*** represents its width. It's given that the rectangle has a length of **7** inches and a width of **6** inches. Substituting **7** for ***ℓ*** and **6** for ***w*** in the formula **$A = \ell w$** yields **$A = (7)(6)$** , or **$A = 42$** . Thus, the area, in square inches, of the rectangle is **42**.

Choice A is incorrect. This is the sum, not the product, of the length and width of the rectangle.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is twice the area, in square inches, of the rectangle.

Question Difficulty:

Easy

Question ID 5252e606

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 5252e606

The side length of a square is **55 centimeters (cm)**. What is the area, **in cm²**, of the square?

- A. 110
- B. 220
- C. 3,025
- D. 12,100

ID: 5252e606 Answer

Correct Answer:

C

Rationale

Choice C is correct. The area **A , in square centimeters (cm²)**, of a square with side length **s , in cm**, is given by the formula **$A = s^2$** . It's given that the square has a side length of **55 cm**. Substituting **55** for **s** in the formula **$A = s^2$** yields **$A = 55^2$** , or **$A = 3,025$** . Therefore, the area, **in cm²**, of the square is **3,025**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the perimeter, **in cm**, of the square, not its area, **in cm²**.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Easy

Question ID bb560789

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: bb560789

Triangle R has an area of **80 square centimeters (cm^2)**. Square S has side lengths of **4 cm**. What is the total area of triangle R and square S, **in cm^2** ?

- A. **42**
- B. **44**
- C. **84**
- D. **96**

ID: bb560789 Answer

Correct Answer:
D

Rationale

Choice D is correct. It's given that triangle R has an area of **80 cm^2** . The area of a square is ℓ^2 , where ℓ is the side length of the square. It's given that square S has side lengths of **4 cm**. It follows that the area, in **cm^2** , of square S is **4^2** , or **16**. Therefore, the total area, in **cm^2** , of triangle R and square S is **$80 + 16$** , or **96**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

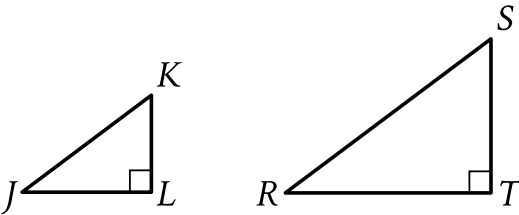
Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy

Question ID babd7461

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: babd7461



Note: Figure not drawn to scale.

In the figure shown, triangle JKL is similar to triangle RST , where J corresponds to R and K corresponds to S . The length of $\overline{JK}$ is 15 , and the perimeter of triangle JKL is 36 . The length of $\overline{RS}$ is 135 . What is the perimeter of triangle RST ?

- A. 324
- B. 540
- C. 2,916
- D. 8,100

ID: babd7461 Answer

Correct Answer:
A

Rationale

Choice A is correct. It's given that triangle JKL is similar to triangle RST , where J corresponds to R and K corresponds to S . It follows that $\overline{JK}$ corresponds to $\overline{RS}$. If two triangles are similar, then the scale factor between their perimeters is equal to the scale factor between the lengths of their corresponding sides. It's given that the length of $\overline{JK}$ is 15 and the length of $\overline{RS}$ is 135 . Therefore, the scale factor from the length of $\overline{JK}$ to the length of $\overline{RS}$ is $\frac{135}{15}$, or 9 . It's given that the perimeter of triangle JKL is 36 . Let p represent the perimeter of triangle RST . It follows that $\frac{p}{36} = 9$. Multiplying each side of this equation by 36 yields $p = 324$. Therefore, the perimeter of triangle RST is 324 .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy

Question ID 59cb654c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 59cb654c

The area of a square is **64** square inches. What is the side length, in inches, of this square?

- A. 8
- B. 16
- C. 64
- D. 128

ID: 59cb654c Answer

Correct Answer:

A

Rationale

Choice A is correct. It's given that the area of a square is **64** square inches. The area **A**, in square inches, of a square is given by the formula $A = s^2$, where **s** is the side length, in inches, of the square. Substituting **64** for **A** in this formula yields $64 = s^2$. Taking the positive square root of both sides of this equation yields $8 = s$. Thus, the side length, in inches, of this square is **8**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the area, in square inches, of the square, not the side length, in inches, of the square.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Easy

Question ID 2b41a4c4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 2b41a4c4

A right rectangular prism has a length of **11** meters, a width of **8** meters, and a height of **10** meters. What is the volume, in cubic meters, of the prism?

ID: 2b41a4c4 Answer

Correct Answer:
880

Rationale

The correct answer is **880**. The volume, V , of a right rectangular prism is given by the formula $V = \ell wh$, where ℓ is the length, w is the width, and h is the height of the prism. It's given that a right rectangular prism has a length of **11** meters, a width of **8** meters, and a height of **10** meters. Substituting **11** for ℓ , **8** for w , and **10** for h in the formula $V = \ell wh$ yields $V = (11)(8)(10)$, or $V = 880$. Therefore, the volume, in cubic meters, of the prism is **880**.

Question Difficulty:
Easy

Question ID d0b6d927

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: d0b6d927

A rectangle has an area of **63** square meters and a length of **9** meters. What is the width, in meters, of the rectangle?

- A. **7**
- B. **54**
- C. **81**
- D. **567**

ID: d0b6d927 Answer

Correct Answer:

A

Rationale

Choice A is correct. The area **A** , in square meters, of a rectangle is the product of its length **ℓ** , in meters, and its width **w** , in meters; thus, **$A = \ell w$** . It's given that a rectangle has an area of **63** square meters and a length of **9** meters. Substituting **63** for **A** and **9** for **ℓ** in the equation **$A = \ell w$** yields **$63 = 9w$** . Dividing both sides of this equation by **9** yields **$7 = w$** . Therefore, the width, in meters, of the rectangle is **7**.

Choice B is incorrect. This is the difference between the area, in square meters, and the length, in meters, of the rectangle, not the width, in meters, of the rectangle.

Choice C is incorrect. This is the square of the length, in meters, not the width, in meters, of the rectangle.

Choice D is incorrect. This is the product of the area, in square meters, and the length, in meters, of the rectangle, not the width, in meters, of the rectangle.

Question Difficulty:

Easy

Question ID 0d43db90

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 0d43db90

The perimeter of triangle ABC is **17** inches, the length of side AB is **4** inches, and the length of side AC is **7** inches. What is the length, in inches, of side BC ?

- A. **4**
- B. **6**
- C. **7**
- D. **11**

ID: 0d43db90 Answer

Correct Answer:
B

Rationale

Choice B is correct. The perimeter of a triangle is the sum of the lengths of all three sides of the triangle. It's given that the lengths of side AB and side AC are **4** inches and **7** inches, respectively. Let x represent the length, in inches, of side BC . The sum of the lengths, in inches, of all three sides of triangle ABC can be represented by the expression $4 + 7 + x$. Since it's given that the perimeter of triangle ABC is **17** inches, it follows that $17 = 4 + 7 + x$, or $17 = 11 + x$. Subtracting **11** from both sides of this equation yields $6 = x$. Therefore, the length, in inches, of side BC is **6**.

Choice A is incorrect. This is the length, in inches, of side AB .

Choice C is incorrect. This is the length, in inches, of side AC .

Choice D is incorrect. This is the sum of the lengths, in inches, of sides AB and AC .

Question Difficulty:
Easy

Question ID 0e40dfb0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 0e40dfb0

A rectangle has a length of **3** units and a width of **39** units. Which expression gives the area, in square units, of this rectangle?

- A. **$2(3 + 39)$**
- B. **$2(3 \cdot 39)$**
- C. **$3 + 39$**
- D. **$3 \cdot 39$**

ID: 0e40dfb0 Answer

Correct Answer:

D

Rationale

Choice D is correct. The area of a rectangle is given by ℓw , where ℓ is the length of the rectangle and w is the width of the rectangle. It's given that a rectangle has a length of **3** units and a width of **39** units. It follows that the area of the rectangle is **$3 \cdot 39$** square units. Therefore, the expression that gives the area, in square units, of this rectangle, is **$3 \cdot 39$** .

Choice A is incorrect. This expression gives the perimeter, in units, of this rectangle.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty:

Easy

Question ID fc5ef8d3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: fc5ef8d3

The table gives the perimeters of similar triangles TUV and XYZ , where $\overline{TU}$ corresponds to $\overline{XY}$. The length of $\overline{TU}$ is 6.

	Perimeter
Triangle TUV	50
Triangle XYZ	150

What is the length of $\overline{XY}$?

- A. 2
- B. 6
- C. 18
- D. 56

ID: fc5ef8d3 Answer

Correct Answer:
C

Rationale

Choice C is correct. It's given that triangle TUV is similar to triangle XYZ , and $\overline{TU}$ corresponds to $\overline{XY}$. If two triangles are similar, then the ratio of their perimeters is equal to the ratio of their corresponding sides. It's given that the perimeter of triangle TUV is 50, the perimeter of triangle XYZ is 150, and the length of $\overline{TU}$ is 6. Let n represent the length of $\overline{XY}$. It follows that $\frac{50}{150} = \frac{6}{n}$, or $\frac{1}{3} = \frac{6}{n}$. Multiplying each side of this equation by n yields $\frac{n}{3} = 6$. Multiplying each side of this equation by 3 yields $n = 18$. Therefore, the length of $\overline{XY}$ is 18.

Choice A is incorrect. This is the solution to $\frac{3}{1} = \frac{6}{n}$, not $\frac{1}{3} = \frac{6}{n}$.

Choice B is incorrect. This is the length of $\overline{TU}$, not $\overline{XY}$.

Choice D is incorrect. This is the sum of the length of $\overline{TU}$ and the perimeter of triangle TUV , not the length of $\overline{XY}$.

Question Difficulty:
Easy

Question ID 689abc2a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 689abc2a

Rectangle P has an area of **72** square inches. If a rectangle with an area of **20** square inches is removed from rectangle P, what is the area, in square inches, of the resulting figure?

- A. **92**
- B. **84**
- C. **80**
- D. **52**

ID: 689abc2a Answer

Correct Answer:
D

Rationale

Choice D is correct. It's given that rectangle P has an area of **72** square inches. If a rectangle with an area of **20** square inches is removed from rectangle P, the area, in square inches, of the resulting figure is **72 — 20**, or **52**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy

Question ID 0837c3b9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 0837c3b9

Triangle ABC and triangle DEF are similar triangles, where $\overline{AB}$ and $\overline{DE}$ are corresponding sides. If $DE = 2AB$ and the perimeter of triangle ABC is 20, what is the perimeter of triangle DEF ?

- A. 10
- B. 40
- C. 80
- D. 120

ID: 0837c3b9 Answer

Correct Answer:
B

Rationale

Choice B is correct. Since triangles ABC and DEF are similar and $DE = 2AB$, the length of each side of triangle DEF is two times the length of its corresponding side in triangle ABC . Therefore, the perimeter of triangle DEF is two times the perimeter of triangle ABC . Since the perimeter of triangle ABC is 20, the perimeter of triangle DEF is 40.

Choice A is incorrect. This is half, not two times, the perimeter of triangle ABC . Choice C is incorrect. This is two times the perimeter of triangle DEF rather than two times the perimeter of triangle ABC . Choice D is incorrect. This is six times, not two times, the perimeter of triangle ABC .

Question Difficulty:
Easy

Question ID 575f1e12

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 575f1e12

What is the area, in square centimeters, of a rectangle with a length of **34 centimeters (cm)** and a width of **29 cm**?

ID: 575f1e12 Answer

Correct Answer:
986

Rationale

The correct answer is **986**. The area, A , of a rectangle is given by $A = \ell w$, where ℓ is the length of the rectangle and w is its width. It's given that the length of the rectangle is **34 centimeters (cm)** and the width is **29 cm**. Substituting **34** for ℓ and **29** for w in the equation $A = \ell w$ yields $A = (34)(29)$, or $A = 986$. Therefore, the area, in square centimeters, of this rectangle is **986**.

Question Difficulty:
Easy

Question ID b4767bef

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: b4767bef

The lengths of two sides of a triangle are **4** centimeters and **6** centimeters. If the perimeter of the triangle is **18** centimeters, what is the length, in centimeters, of the third side of this triangle?

- A. **2**
- B. **8**
- C. **10**
- D. **24**

ID: b4767bef Answer

Correct Answer:
B

Rationale

Choice B is correct. The perimeter of a triangle is the sum of the lengths of all three of its sides. It's given that the lengths of two sides of a triangle are **4** centimeters and **6** centimeters. Let x represent the length, in centimeters, of the third side of this triangle. The sum of the lengths, in centimeters, of all three sides of the triangle can be represented by the expression $4 + 6 + x$. Since it's given that the perimeter of the triangle is **18** centimeters, it follows that $4 + 6 + x = 18$, or $10 + x = 18$. Subtracting **10** from both sides of this equation yields $x = 8$. Therefore, the length, in centimeters, of the third side of this triangle is **8**.

Choice A is incorrect. If the length of the third side of this triangle were **2** centimeters, the perimeter, in centimeters, of the triangle would be $4 + 6 + 2$, or **12**, not **18**.

Choice C is incorrect. If the length of the third side of this triangle were **10** centimeters, the perimeter, in centimeters, of the triangle would be $4 + 6 + 10$, or **20**, not **18**.

Choice D is incorrect. If the length of the third side of this triangle were **24** centimeters, the perimeter, in centimeters, of the triangle would be $4 + 6 + 24$, or **34**, not **18**.

Question Difficulty:
Easy

Question ID c88183f7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: c88183f7

A rectangle has a length of **13** and a width of **6**. What is the perimeter of the rectangle?

- A. **12**
- B. **26**
- C. **38**
- D. **52**

ID: c88183f7 Answer

Correct Answer:

C

Rationale

Choice C is correct. The perimeter of a quadrilateral is the sum of the lengths of its four sides. It's given that the rectangle has a length of **13** and a width of **6**. It follows that the rectangle has two sides with length **13** and two sides with length **6**. Therefore, the perimeter of the rectangle is **13 + 13 + 6 + 6**, or **38**.

Choice A is incorrect. This is the sum of the lengths of the two sides with length **6**, not the sum of the lengths of all four sides of the rectangle.

Choice B is incorrect. This is the sum of the lengths of the two sides with length **13**, not the sum of the lengths of all four sides of the rectangle.

Choice D is incorrect. This is the perimeter of a rectangle that has four sides with length **13**, not two sides with length **13** and two sides with length **6**.

Question Difficulty:

Easy

Question ID 50774285

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 50774285

Each base of a right rectangular prism has a length of **19** inches and a width of **8** inches. The prism has a volume of **2,736** cubic inches. What is the height, in inches, of the prism?

- A. **18**
- B. **27**
- C. **144**
- D. **152**

ID: 50774285 Answer

Correct Answer:
A

Rationale

Choice A is correct. The volume, V , of a rectangular prism is given by the formula $V = \ell wh$, where ℓ is the length of the base, w is the width of the base, and h is the height of the prism. It's given that each base of a right rectangular prism has a length of **19** inches and a width of **8** inches, and that the prism has a volume of **2,736** cubic inches. Substituting **19** for ℓ , **8** for w , and **2,736** for V in the formula $V = \ell wh$ gives $2,736 = (19)(8)(h)$, or $2,736 = 152h$. Dividing each side of this equation by **152** yields $18 = h$. Therefore, the height, in inches, of the prism is **18**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

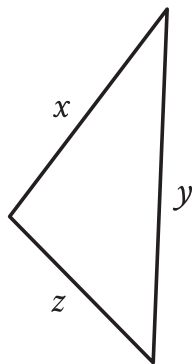
Choice D is incorrect. This is the area, in square inches, of the base of the prism, not the height, in inches, of the prism.

Question Difficulty:
Easy

Question ID 29e9b28c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 29e9b28c



Note: Figure not drawn to scale.

The triangle shown has a perimeter of **22** units. If $x = 9$ units and $y = 7$ units, what is the value of z , in units?

- A. **6**
- B. **7**
- C. **9**
- D. **16**

ID: 29e9b28c Answer

Correct Answer:

A

Rationale

Choice A is correct. The perimeter of a triangle is the sum of the lengths of its three sides. The triangle shown has side lengths x , y , and z . It's given that the triangle has a perimeter of **22** units. Therefore, $x + y + z = 22$. If $x = 9$ units and $y = 7$ units, the value of z , in units, can be found by substituting **9** for x and **7** for y in the equation $x + y + z = 22$, which yields $9 + 7 + z = 22$, or $16 + z = 22$. Subtracting **16** from both sides of this equation yields $z = 6$. Therefore, if $x = 9$ units and $y = 7$ units, the value of z , in units, is **6**.

Choice B is incorrect. This is the value of y , in units, not the value of z , in units.

Choice C is incorrect. This is the value of x , in units, not the value of z , in units.

Choice D is incorrect. This is the value of $x + y$, in units, not the value of z , in units.

Question Difficulty:

Easy

Question ID 3453aafc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 3453aafc

What is the area, in square centimeters, of a rectangle with a length of **36** centimeters and a width of **34** centimeters?

- A. **70**
- B. **140**
- C. **1,156**
- D. **1,224**

ID: 3453aafc Answer

Correct Answer:

D

Rationale

Choice D is correct. The area **A** , in square centimeters, of a rectangle can be found using the formula **$A = \ell w$** , where ℓ is the length, in centimeters, of the rectangle and w is its width, in centimeters. It's given that the rectangle has a length of **36** centimeters and a width of **34** centimeters. Substituting **36** for ℓ and **34** for w in the formula **$A = \ell w$** yields **$A = 36(34)$** , or **$A = 1,224$** . Therefore, the area, in square centimeters, of this rectangle is **1,224**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the perimeter, in centimeters, not the area, in square centimeters, of the rectangle.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Easy

Question ID f60bb551

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: f60bb551

The area of a rectangle is **630** square inches. The length of the rectangle is **70** inches. What is the width, in inches, of this rectangle?

- A. **9**
- B. **70**
- C. **315**
- D. **560**

ID: f60bb551 Answer

Correct Answer:
A

Rationale

Choice A is correct. The area **A** , in square inches, of a rectangle is the product of its length **ℓ** , in inches, and its width **w** , in inches; thus, **$A = \ell w$** . It's given that the area of a rectangle is **630** square inches and the length of the rectangle is **70** inches. Substituting **630** for **A** and **70** for **ℓ** in the equation **$A = \ell w$** yields **$630 = 70w$** . Dividing both sides of this equation by **70** yields **$9 = w$** . Therefore, the width, in inches, of this rectangle is **9**.

Choice B is incorrect. This is the length, not the width, in inches, of the rectangle.

Choice C is incorrect. This is half the area, in square inches, not the width, in inches, of the rectangle.

Choice D is incorrect. This is the difference between the area, in square inches, and the length, in inches, of the rectangle, not the width, in inches, of the rectangle.

Question Difficulty:
Easy

Question ID d2047497

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: d2047497

What is the area of a rectangle with a length of **17 centimeters (cm)** and a width of **7 cm**?

- A. **24 cm²**
- B. **48 cm²**
- C. **119 cm²**
- D. **576 cm²**

ID: d2047497 Answer

Correct Answer:

C

Rationale

Choice C is correct. The area of a rectangle with length *l* and width *w* can be found using the formula $A = lw$. It's given that the rectangle has a length of **17 cm** and a width of **7 cm**. Therefore, the area of this rectangle is $A = 17(7)$, or **119 cm²**.

Choice A is incorrect. This is the sum of the length and width of the rectangle, not the area.

Choice B is incorrect. This is the perimeter of the rectangle, not the area.

Choice D is incorrect. This is the sum of the length and width of the rectangle squared, not the area.

Question Difficulty:

Easy

Question ID 4efea6a3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 4efea6a3

The area of a rectangle is **57** square inches. The length of the longest side of the rectangle is **19** inches. What is the length, in inches, of the shortest side of this rectangle?

ID: 4efea6a3 Answer

Correct Answer:

3

Rationale

The correct answer is **3**. The area of a rectangle can be calculated by multiplying the length of its longest side by the length of its shortest side. It's given that the area of the rectangle is **57** square inches and the length of the longest side of the rectangle is **19** inches. Let x represent the length, in inches, of the shortest side of this rectangle. It follows that $57 = 19x$. Dividing both sides of this equation by **19** yields $3 = x$. Therefore, the length, in inches, of the shortest side of the rectangle is **3**.

Question Difficulty:

Easy

Question ID f39f88b7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: f39f88b7

A triangle has a base length of **40** centimeters and a height of **90** centimeters. What is the area, in square centimeters, of the triangle?

ID: f39f88b7 Answer

Correct Answer:
1800

Rationale

The correct answer is **1,800**. The area, A , of a triangle can be found using the formula $A = \frac{1}{2}bh$, where b is the base length of the triangle and h is the height of the triangle. It's given that the triangle has a base length of **40** centimeters and a height of **90** centimeters. Substituting **40** for b and **90** for h in the formula $A = \frac{1}{2}bh$ yields $A = \frac{1}{2}(40)(90)$, or $A = 1,800$. Therefore, the area, in square centimeters, of the triangle is **1,800**.

Question Difficulty:
Easy

Question ID 76670c80

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 76670c80

Each side of a square has a length of **45**. What is the perimeter of this square?

ID: 76670c80 Answer

Correct Answer:

180

Rationale

The correct answer is **180**. The perimeter of a polygon is equal to the sum of the lengths of the sides of the polygon. It's given that each side of the square has a length of **45**. Since a square is a polygon with **4** sides, the perimeter of this square is **$45 + 45 + 45 + 45$** , or **180**.

Question Difficulty:

Easy

Question ID 4420e500

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 4420e500

What is the area of a rectangle with a length of **4 centimeters (cm)** and a width of **2 cm**?

- A. **6 cm²**
- B. **8 cm²**
- C. **12 cm²**
- D. **36 cm²**

ID: 4420e500 Answer

Correct Answer:

B

Rationale

Choice B is correct. The area of a rectangle with length ℓ and width w can be found using the formula $A = \ell w$. It's given that the rectangle has a length of **4 cm** and a width of **2 cm**. Therefore, the area of this rectangle is **(4 cm)(2 cm)**, or **8 cm²**.

Choice A is incorrect. This is the sum, **in cm**, of the length and width of the rectangle, not the area, **in cm²**.

Choice C is incorrect. This is the perimeter, **in cm**, of the rectangle, not the area, **in cm²**.

Choice D is incorrect. This is the sum of the length and width of the rectangle squared, not the area.

Question Difficulty:

Easy

Question ID 165c30c4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 165c30c4

A rectangle has a length of **64** inches and a width of **32** inches. What is the area, in square inches, of the rectangle?

ID: 165c30c4 Answer

Correct Answer:
2048

Rationale

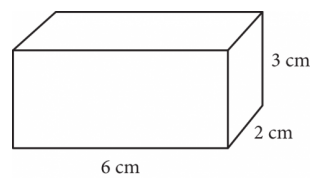
The correct answer is **2,048**. The area **A** , in square inches, of a rectangle is equal to the product of its length **ℓ** , in inches, and its width **w** , in inches, or **$A = \ell w$** . It's given that the rectangle has a length of **64** inches and a width of **32** inches. Substituting **64** for **ℓ** and **32** for **w** in the equation **$A = \ell w$** yields **$A = (64)(32)$** , or **$A = 2,048$** . Therefore, the area, in square inches, of the rectangle is **2,048**.

Question Difficulty:
Easy

Question ID d683a9cc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: d683a9cc



The figure shows the lengths, in centimeters (cm), of the edges of a right rectangular prism. The volume V of a right rectangular prism is ℓwh , where ℓ is the length of the prism, w is the width of the prism, and h is the height of the prism. What is the volume, in cubic centimeters, of the prism?

- A. 36
- B. 24
- C. 12
- D. 11

ID: d683a9cc Answer

Correct Answer:
A

Rationale

Choice A is correct. It’s given that the volume of a right rectangular prism is ℓwh . The prism shown has a length of 6 cm, a width of 2 cm, and a height of 3 cm. Thus, $\ell wh = (6)(2)(3)$, or 36 cubic centimeters.

Choice B is incorrect. This is the volume of a rectangular prism with edge lengths of 6, 2, and 2. Choice C is incorrect and may result from only finding the product of the length and width of the base of the prism. Choice D is incorrect and may result from finding the sum, not the product, of the edge lengths of the prism.

Question Difficulty:
Easy

Question ID a490003a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: a490003a

The width of a rectangle is **7** centimeters. The length of the rectangle is **40** centimeters longer than the width. What is the area, in square centimeters, of this rectangle?

- A. **7**
- B. **14**
- C. **54**
- D. **329**

ID: a490003a Answer

Correct Answer:
D

Rationale

Choice D is correct. It’s given that the width of this rectangle is **7** centimeters and that the length of this rectangle is **40** centimeters longer than the width. Therefore, the length of this rectangle is **7 + 40**, or **47**, centimeters. The area of a rectangle can be found by multiplying its length and its width. Therefore the area, in square centimeters, of this rectangle is **(7)(47)**, or **329**.

Choice A is incorrect. This is the width, in centimeters, not the area, in square centimeters, of this rectangle.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy

Question ID 02b02213

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 02b02213

What is the perimeter, in inches, of a rectangle with a length of **4** inches and a width of **9** inches?

- A. **13**
- B. **17**
- C. **22**
- D. **26**

ID: 02b02213 Answer

Correct Answer:

D

Rationale

Choice D is correct. The perimeter of a figure is equal to the sum of the measurements of the sides of the figure. It's given that the rectangle has a length of **4** inches and a width of **9** inches. Since a rectangle has **4** sides, of which opposite sides are parallel and equal, it follows that the rectangle has two sides with a length of **4** inches and two sides with a width of **9** inches. Therefore, the perimeter of this rectangle is **4 + 4 + 9 + 9**, or **26** inches.

Choice A is incorrect. This is the sum, in inches, of the length and the width of the rectangle.

Choice B is incorrect. This is the sum, in inches, of the two lengths and the width of the rectangle.

Choice C is incorrect. This is the sum, in inches, of the length and the two widths of the rectangle.

Question Difficulty:

Easy